

Credit Spread Risk in the Banking Book (CSRBB)

Framework Summary: Swedish Bond vs German Bund (EUR)

Trade Republic — Risk Modelling

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Contents

1	Rationale of the Approach	2
1.1	Regulatory Context	2
1.2	Instrument Pair: Swedish Bond vs German Bund	2
1.3	Choice of Risk-Free Curve	2
2	Mathematical Theory	2
2.1	Zero Coupon Bond and Spot Rate	2
2.2	Forward Rate	3
2.3	Nelson-Siegel-Svensson (NSS) Yield Curve	3
2.4	Credit Spread	3
2.5	Value at Risk	3
2.6	Expected Shortfall	4
2.7	Economic Value of Equity (EVE)	4
2.8	Net Interest Income (NII)	4
2.9	Capital Requirement	5
3	Implementation Guidelines	5
3.1	Step-by-Step Procedure	5
3.2	Key Modelling Decisions	5
3.3	Conservative Adjustments	6
4	Backtesting Guidelines	6
4.1	Objective	6
4.2	Procedure	6
4.3	Statistical Assessment — Binomial Test	6
4.4	Basel Traffic-Light Framework	6
4.5	Validation Frequency	7

1 Rationale of the Approach

1.1 Regulatory Context

The European Banking Authority (EBA) updated its guidelines on Interest Rate Risk and Credit Spread Risk in the Banking Book via **EBA/GL/2022/14**, effective 31 December 2023. The framework elevates CSRBB to a *standalone* risk category, distinct from IRRBB, and mandates:

- Assessment of CSRBB across all credit-spread-sensitive instruments (assets, liabilities, off-balance-sheet);
- Inclusion of all fair-valued assets at a minimum;
- Measurement in terms of both **EVE** (Economic Value of Equity) and **NII** (Net Interest Income);
- Justified exclusion of any instrument from scope.

In Sweden, Finansinspektionen (FI) additionally recommends a VaR-based internal model using a **90-day holding period** and a **99% confidence level**.

1.2 Instrument Pair: Swedish Bond vs German Bund

The credit spread is defined as the yield differential between a risky instrument and a risk-free benchmark of the same maturity:

$$CS(t, m) = R_{\text{risky}}(t, m) - R_{\text{rf}}(t, m)$$

In the EUR context adopted here:

Component	Instrument
Risk-free benchmark	German Bund (Bundesanleihe)
Risky instrument	Swedish EUR-denominated bond (covered / bank / sovereign)

The spread embeds two components per EBA (2022):

1. **Market credit spread:** compensation for counterparty default risk at a given credit quality;
2. **Market liquidity spread:** compensation for the lower liquidity of the Swedish instrument relative to the Bund.

1.3 Choice of Risk-Free Curve

The ECB publishes AAA-rated EUR sovereign yield curves daily, which serve as the Bund-equivalent risk-free reference. This avoids the issue encountered when using OIS curves for portfolios with significant floating-rate exposure: the credit spread would collapse to zero at short maturities if Stibor/EURIBOR were used both as the benchmark and the underlying floating rate.

2 Mathematical Theory

2.1 Zero Coupon Bond and Spot Rate

A Zero Coupon Bond (ZCB) with maturity T is priced at time t as:

$$p(t, T) = e^{-r(t, T)(T-t)}$$

where $r(t, T)$ is the continuously compounded spot rate.

2.2 Forward Rate

The forward rate $F(t; S, T)$ agreed at time t for the period $[S, T]$ is:

$$F(t; S, T) = -\frac{\ln p(t, T) - \ln p(t, S)}{T - S}$$

For adjacent time bucket midpoints T_k and T_{k+1} , the effective forward rate at time $t = 0$ is:

$$F_0(0; T_k, T_{k+1}) = \frac{r(0, T_{k+1}) T_{k+1} - r(0, T_k) T_k}{T_{k+1} - T_k}$$

2.3 Nelson-Siegel-Svensson (NSS) Yield Curve

Since market data is discrete and often incomplete for longer maturities, the NSS model fits a smooth yield curve. The instantaneous forward rate is:

$$r(t, T) = \beta_0 + \beta_1 e^{-T/\tau_1} + \beta_2 \left(\frac{T}{\tau_1} \right) e^{-T/\tau_1} + \beta_3 \left(\frac{T}{\tau_2} \right) e^{-T/\tau_2}$$

The zero yield curve is:

$$R(t, T) = \beta_0 + \beta_1 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} \right) + \beta_2 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} - e^{-T/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2} \right)$$

Parameters $\{\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2\}$ are calibrated daily via non-linear least squares:

$$\min_{\lambda, \beta} \sum_T [\hat{r}(t, T) - r^M(t, T)]^2 \quad \forall t$$

where $\hat{r}(t, T)$ is the model estimate and $r^M(t, T)$ is the observed market rate.

2.4 Credit Spread

After constructing the daily NSS yield curves for both the risky and risk-free instruments, the historical credit spread for each day t and maturity m is:

$$CS(t, m) = R_{\text{risky}}(t, m) - R_{\text{rf}}(t, m)$$

2.5 Value at Risk

For a portfolio with future value X at time 1 and discounted loss $L = -X/R_0$, the VaR at confidence level $1 - p$ is the $(1 - p)$ -quantile of L :

$$\text{VaR}_p(X) = F_L^{-1}(1 - p)$$

The Historical Simulation VaR sorts all historical 90-day changes in descending order and takes the 99th percentile as the capital requirement.

2.6 Expected Shortfall

Expected Shortfall (ES) is a coherent risk measure that averages losses beyond the VaR cutoff:

$$ES_p(X) = \frac{1}{p} \int_0^p \text{VaR}_u(X) du = \mathbb{E}[L \mid L \geq \text{VaR}_p(X)]$$

ES satisfies subadditivity (portfolio diversification reduces risk), which VaR does not.

2.7 Economic Value of Equity (EVE)

The discount factor for currency c , time bucket i with maturity m_i , at time t is:

$$DF_{i,c,t} = e^{-R_{i,c}(t, m_i) \cdot (m_i - t_0)}$$

The EVE under the stressed scenario (risky yield as discount rate) is:

$$EVE_{\text{stressed}, c, t} = \sum_{i=1}^{19} CF_{i,c} \cdot DF_{i,c,t}$$

The change in EVE due to credit spread risk is:

$$\Delta EVE_{c,t} = EVE_{\text{stressed}, c, t} - EVE_{\text{base}, c, t}$$

where the base uses the German Bund yield curve as the discount rate. Aggregating across currencies (converted to a common base):

$$\Delta EVE_t = \sum_c \Delta EVE_{c,t}$$

The 90-day rolling CSRBB capital requirement:

$$\text{VaR}_{0.99}(\Delta EVE^{90}) = \min \{m : P(\Delta EVE^{90} \leq m) \leq 0.01\}$$

2.8 Net Interest Income (NII)

The change in NII for currency c at time t , driven by credit spread movements between time bucket midpoints T_k and T_{k+1} , is:

$$\Delta NII_{c,t} = \sum_{i=1}^{19} CF_{i,c} \cdot \frac{CS_t(0; T_{k+1}) T_{k+1} - CS_t(0; T_k) T_k}{T_{k+1} - T_k}$$

Aggregating across currencies:

$$\Delta NII_t = \sum_c \Delta NII_{c,t}$$

The 90-day rolling NII capital requirement:

$$\text{VaR}_{0.99}(\Delta NII^{90}) = \min \{m : P(\Delta NII^{90} \leq m) \leq 0.01\}$$

2.9 Capital Requirement

EVE and NII respond to credit spread movements in opposite directions: EVE is sensitive to *widening* spreads; NII (for an asset-heavy book) is sensitive to *tightening* spreads. The capital requirement is therefore:

$$K_{\text{CSRBB}} = \max(\text{VaR}_{0.99}(\Delta EVE^{90}), \text{VaR}_{0.99}(\Delta NII^{90}))$$

3 Implementation Guidelines

3.1 Step-by-Step Procedure

1. **Scope identification:** identify all credit-spread-sensitive instruments on the banking book (assets, liabilities, off-balance-sheet). For EUR-denominated Swedish bonds, include fixed-coupon, zero-coupon, and floating-rate instruments linked to EURIBOR.
2. **Cash flow bucketing:** slot all net notional repricing cash flows $CF_{i,c}$ into the 19 BCBS time buckets. Floating-rate instruments are treated as fixed until the next repricing date.
3. **Data sourcing:** retrieve at least **10 years** of daily yield data to ensure coverage of multiple stress periods (e.g., Eurozone debt crisis 2012, COVID-19 2020, Ukraine shock 2022). Use ECB AAA sovereign yield curves for the German Bund risk-free curve.
4. **Yield curve fitting:** apply the NSS model daily to both the Swedish bond and German Bund yield data. Validate the resulting curves visually to ensure stability up to 25-year maturities.
5. **Credit spread time series:** compute daily $CS(t, m)$ for each time bucket maturity m .
6. **EVE and NII calculation:** compute $\Delta EVE_{c,t}$ and $\Delta NII_{c,t}$ for each currency and day. Convert to a common currency and aggregate.
7. **90-day rolling differences:** compute ΔEVE^{90} and ΔNII^{90} as rolling 90-day differences over the full historical series.
8. **Risk metrics:** sort ΔEVE^{90} and ΔNII^{90} in descending order and extract $\text{VaR}_{0.99}$ and $\text{ES}_{0.99}$.
9. **Capital requirement:** take the maximum of the two VaR measures as the CSRBB capital requirement.

3.2 Key Modelling Decisions

Decision	Recommendation
Risk-free curve	ECB AAA EUR sovereign (Bund-equivalent); avoid EURIBOR OIS for EURIBOR-linked floating instruments
VaR approach	Portfolio-level HSVaR (vs instrument-level): more realistic, avoids combining independent stress events
Liabilities in EVE	Exclude unless callable / re-emittable; include floating-rate liabilities in NII
Mirroring	Optional; significantly increases conservatism — use as a sensitivity check
Data window	Minimum 10 years; extend to 15+ for greater stress coverage

3.3 Conservative Adjustments

- **Instrument-level VaR:** compute VaR separately per instrument, then sum — allows independent stress scenarios across instruments.
- **Mirroring:** for each observed CS movement ΔCS , include $-\Delta CS$ in the dataset, doubling the sample and forcing symmetry.
- **Extended data window:** include pre-2013 GFC data to capture the 2008–2009 sovereign spread crisis.

4 Backtesting Guidelines

4.1 Objective

Backtesting verifies that the $\text{VaR}_{0.99}$ model is correctly calibrated: the actual 90-day loss should exceed the predicted VaR in no more than 1% of observations.

4.2 Procedure

1. For each trading day t in the backtesting window (full calendar year recommended), compute the VaR and ES using data up to day t .
2. Record the actual realised ΔEVE_t^{90} and ΔNII_t^{90} .
3. Flag an **outlier** (exception) whenever the realised value exceeds the predicted VaR.
4. Count the total number of outliers x over N observations.

4.3 Statistical Assessment — Binomial Test

The probability of observing exactly x outliers out of N days, under the null hypothesis $H_0 : p \leq p_0 = 0.01$, follows a binomial distribution:

$$P(X = x \mid N, p_0) = \binom{N}{x} p_0^x (1 - p_0)^{N-x}$$

The cumulative probability of observing at most x outliers:

$$P(X \leq x \mid N, p_0) = \sum_{k=0}^x \binom{N}{k} p_0^k (1 - p_0)^{N-k}$$

4.4 Basel Traffic-Light Framework

The Basel II Annex 10a “three-zone” approach classifies backtesting results as follows (for $N \approx 2600$ days, $p_0 = 0.01$):

Zone	Outlier Count	Interpretation
Green	$0 \leq x \leq 28$	Model acceptable
Yellow	$29 \leq x \leq 38$	Model under review
Red	$x \geq 39$	Model likely rejected

For a 1-year backtesting window ($N \approx 250$ trading days):

$$\text{Expected outliers} = N \times p_0 = 250 \times 0.01 = 2.5$$

4.5 Validation Frequency

- Backtest against the **full previous year** at each annual model review.
- Monitor rolling outlier counts quarterly to detect model drift.
- Re-calibrate NSS parameters and data windows whenever the model enters the Yellow zone.
- Document all modelling choices and deviations from EBA guidelines in line with EBA/GL/2022/14.

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